CS101: Foundations of Computer Science

Academic Term: Fall 2026 | **Instructor:** Dr. Eleanor Vance | **Dates:** Sep 07 - Dec 18, 2026

Course Overview

This course provides a rigorous introduction to computational thinking, algorithms, and data structures. Students will master control flow, functions, modular architecture, recursion, and object-oriented design.

Grading Breakdown & Assessment Schedule

Assessment Name	Type	Due Date	Weight	Topics Covered
Assignment 1: Logic & Flow	Assignment	2026-09-25	10%	Variables, Loops, Functions
Assignment 2: Collections	Assignment	2026-10-09	10%	Lists, Dictionaries, Sets
Midterm Examination 1	Midterm	2026-10-15	15%	Recursion, Complexity, Flow
Assignment 3: OOP Architecture	Assignment	2026-11-06	10%	Classes, Inheritance, Mixins
Midterm Examination 2	Midterm	2026-11-19	15%	Trees, Sorting, Binary Search
Capstone Software Project	Project	2026-12-04	15%	End-to-End System Design
Final Comprehensive Examination	Final	2026-12-16	25%	Comprehensive All Modules

Weekly Modules & Syllabus Topics

- Week 1-2: Primitive Types, Expressions, Functions & Branching
- Week 3-4: Strings, Lists, Dictionaries, and File I/O
- Week 5-6: Algorithmic Complexity (Big-O) and Recursion
- Week 7-8: Object-Oriented Programming (Classes, Inheritance, Encapsulation)
- Week 9-10: Elementary Data Structures (Stacks, Queues, Linked Lists, Trees)
- Week 11-12: Searching and Sorting (QuickSort, MergeSort, Binary Search)
- Week 13-14: Software Engineering Principles and Capstone Project Integration
- Week 15: Final Review & Exam Preparation